R. file × | R. file 2 ×

Run | Source

```r
# BIOL 1002 R Assignment
# Name: Danika McNeill
# Date: February 11
# Install packages
install.packages("dplyr")
# BIOL 1002 R Assignment
# Name: Danika McNeill
# Date: February 11, 2026
# PART I: Moose Populations in Newfoundland
# Install packages
install.packages("dplyr")
# Question 1: Load dplyr
library(dplyr)
# Question 2: Import data
moosedata <- read.csv("MoosePopulation.csv")
# Question 3: Remove missing values
moose_clean <- na.omit(moosedata)
# Question 4: Select specific columns
moose_sel <- select(moose_clean, Ecoregion, Year, Area, Estimated_Moose_Pop)
# Question 5:
# a) Oldest observation
year_min <- min(moose_sel$Year)
# b) Highest moose population
moose_max <- max(moose_sel$Estimated_Moose_Pop, na.rm = TRUE)
# Question 6: Calculate moose density
moosedata2 <- mutate(moose_sel, MooseDensity = Estimated_Moose_Pop / Area)
# Question 7: Line graph of moose density over time
plot(moosedata2$Year, moosedata2$MooseDensity,
type = "l",
xlab = "Year",
ylab = "Moose per sq km",
main = "Moose density in Newfoundland ecoregions over time")
# Question 8:
# a) Filter for Western_Forests
moose_west <- filter(moosedata2, Ecoregion == "Western_Forests")
# b) Line graph for Western Forests
plot(moose_west$Year, moose_west$MooseDensity,
type = "l",
xlab = "Year",
ylab = "Moose per sq km",
main = "Moose density in Western Forests over time")
# Question 9:
# a) Filter for year 2020
moose_2020 <- filter(moosedata2, Year == 2020)
# b) Filter for high density (> 2.0)
moose_2020_high <- filter(moose_2020, MooseDensity > 2.0)
# c) Arrange in descending order
moose_2020_high_by0 <- arrange(moose_2020_high, desc(MooseDensity))
# Question 10: Use pipes for same operations
moosefinal <- moosedata2 %>%
filter(Year == 2020) %>%
filter(MooseDensity > 2.0) %>%
arrange(desc(MooseDensity)) %>%
print()
# PART II: Tree Sapling Study
# Question 11:
# a) Load sapling data
saplings <- read.csv("SaplingStudy.csv")
# b) Remove NAs
sap_clean <- na.omit(saplings)
# Question 12:
# a) Average browsing by ecoregion
sap_reg_browse <- sap_clean %>%
group_by(Ecoregion) %>%
summarize(AverageBrowsing = mean(BrowsingScore)) %>%
print()
# b) Arrange and comment
avg_browse_reg <- arrange(sap_reg_browse, desc(AverageBrowsing))
# Western_Forests has highest browsing (5.0), Long_Range_Barrens has lowest (2.6)
# Question 13:
# a) Average height by ecoregion
sap_reg_height <- sap_clean %>%
group_by(Ecoregion) %>%
summarize(AverageHeight = mean(Height)) %>%
print()
# b) Filter for low heights
sap_reg_height_low <- filter(sap_reg_height, AverageHeight < 20)
print(sap_reg_height_low)
# No ecoregions have average height less than 20 cm
# Question 14:
# a) Average browsing by species
sap_spe_browse <- sap_clean %>%
group_by(Species) %>%
summarize(AverageBrowsing = mean(BrowsingScore)) %>%
print()
# b) Arrange and comment
avg_browse_spe <- arrange(sap_spe_browse, desc(AverageBrowsing))
# Black_Ash has highest browsing (5.0), Black_Spruce has lowest (2.4)
# Question 15: Balsam Fir browsing by ecoregion
fir_reg_browse <- sap_clean %>%
filter(Species == "Balsam_Fir") %>%
group_by(Ecoregion) %>%
summarize(AverageBrowsing = mean(BrowsingScore)) %>%
print()
# Question 16: Bar graph for Balsam Fir
barplot(fir_reg_browse$AverageBrowsing,
names.arg = fir_reg_browse$Ecoregion,
xlab = "Ecoregion",
ylab = "Average Browsing Score",
main = "Balsam Fir Browsing by Ecoregion",
col = "forestgreen",
cex.names = 0.6)
# Question 17:
# a) Black Spruce browsing by ecoregion
spruce_reg_browse <- sap_clean %>%
filter(Species == "Black_Spruce") %>%
group_by(Ecoregion) %>%
summarize(AverageBrowsing = mean(BrowsingScore)) %>%
print()
# b) Bar graph for Black Spruce
barplot(spruce_reg_browse$AverageBrowsing,
names.arg = spruce_reg_browse$Ecoregion,
xlab = "Ecoregion",
ylab = "Average Browsing Score",
main = "Black Spruce Browsing by Ecoregion",
col = "darkblue",
cex.names = 0.6)
# c) Comparison comment
# Black Spruce generally has lower browsing scores than Balsam Fir across ecoregions
# Question 18: Count trees by ecoregion
sap_reg_tally <- sap_clean %>%
group_by(Ecoregion) %>%
tally() %>%
print()
print()
# Question 16: Bar graph for Balsam Fir
barplot(fir_reg_browse$AverageBrowsing,
names.arg = fir_reg_browse$Ecoregion,
xlab = "Ecoregion",
ylab = "Average Browsing Score",
main = "Balsam Fir Browsing by Ecoregion",
col = "forestgreen",
cex.names = 0.6)
# Question 17:
# a) Black Spruce browsing by ecoregion
spruce_reg_browse <- sap_clean %>%
filter(Species == "Black_Spruce") %>%
group_by(Ecoregion) %>%
summarize(AverageBrowsing = mean(BrowsingScore)) %>%
print()
# b) Bar graph for Black Spruce
barplot(spruce_reg_browse$AverageBrowsing,
names.arg = spruce_reg_browse$Ecoregion,
xlab = "Ecoregion",
ylab = "Average Browsing Score",
main = "Black Spruce Browsing by Ecoregion",
col = "darkblue",
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# Question 18: Count trees by ecoregion
sap_reg_tally <- sap_clean %>%
group_by(Ecoregion) %>%
tally() %>%
print()
# Question 19: Count trees by species
sap_spe_tally <- sap_clean %>%
group_by(Species) %>%
tally() %>%
print()
# Question 20:
# a) Sampling bias comment
# The dataset is not evenly distributed. North_Shore_Forests has 8 samples while some regions have on
# b) Importance of recognizing bias
# Recognizing bias is important because uneven sampling can lead to incorrect conclusions about speci
# PART III: Creating and Joining Datasets
# Question 21:
# a) Prepare moose data for 2020
moose_2020b <- moose_clean %>%
filter(Year == 2020) %>%
mutate(MooseDensity = Estimated_Moose_Pop / Area)
# b) Join datasets
moose_sap <- left_join(moose_2020b, sap_clean, by = 'Ecoregion')
# Question 22: Average browsing and density by species and ecoregion
sum_spe_browse <- moose_sap %>%
group_by(Species, Ecoregion) %>%
summarize(AvgBrowsing = mean(BrowsingScore),
AvgDensity = mean(MooseDensity)) %>%
print()
# Question 23:
# a) Hypothesis comment
# The hypothesis is not strongly supported - moose show consistent preferences across different densi
# b) Species preference comment
# Moose seem to favor Black_Ash and Willow most, and browse Black_Spruce the least.
# c) Missing species comment
# Black_Ash is not shown on the figure because it only appears in one ecoregion.
# Question 24: Create collisions dataset
collisions2020 <- c(56, 60, 14, 36, 48, 10, 40, 110, 6)
human_pop <- c(18000, 12000, 4000, 75100, 24000, 3500, 32000, 270000, 2300)
study_sites <- c("North_Shore_Forests","Northern_Peninsula_Forests", "Long_Range_Barrens","Central_Fo
moose_coll <- data.frame(collisions2020, human_pop, study_sites)
# Question 25:
# a) Rename column
moose_coll2 <- rename(moose_coll, Ecoregion = study_sites)
# b) Join datasets
coll_merge <- left_join(moose_2020, moose_coll2, by = "Ecoregion")
# Question 26:
# a) Scatterplot
plot(coll_merge$MooseDensity, coll_merge$collisions2020,
xlab = "Moose Density",
ylab = "Number of Collisions",
main = "Moose-Vehicle Collisions vs Moose Density")
# b) Trend comment
# There appears to be a positive trend: higher moose density correlates with more collisions. Avalon_
# Question 27: Calculate collisions per capita
coll_merge_per_capita <- coll_merge %>%
mutate(coll_per_capita = collisions2020 / human_pop)
# Question 28: Scatterplot of collisions per capita
plot(coll_merge_per_capita$human_pop, coll_merge_per_capita$coll_per_capita,
xlab = "Human Population",
ylab = "Collisions per Capita",
main = "Moose Collisions per Person by Population Size")
# Question 29: Trend comment
# Collisions per capita decrease as human population increases, which makes sense because larger popu
savehistory("/cloud/project/R. Code")
```
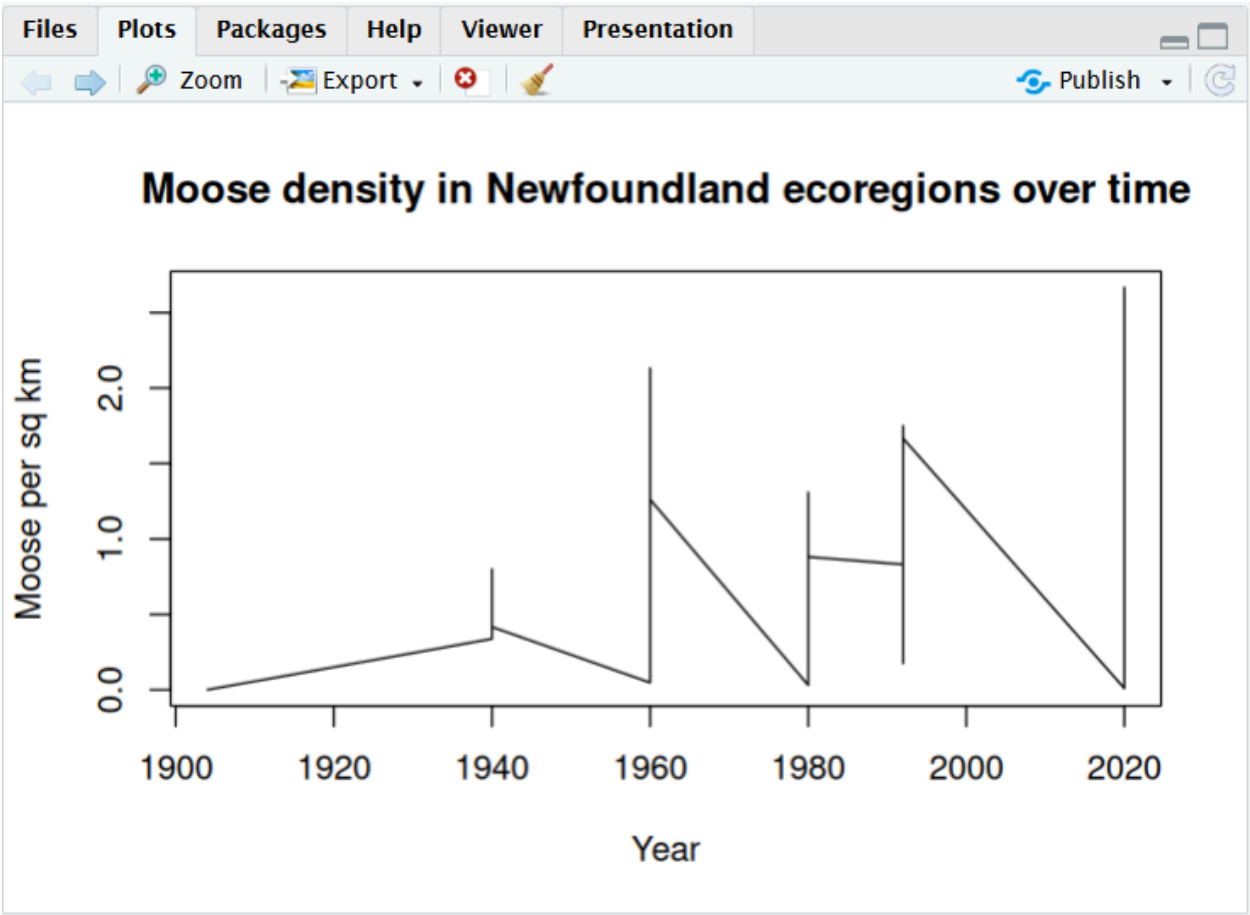
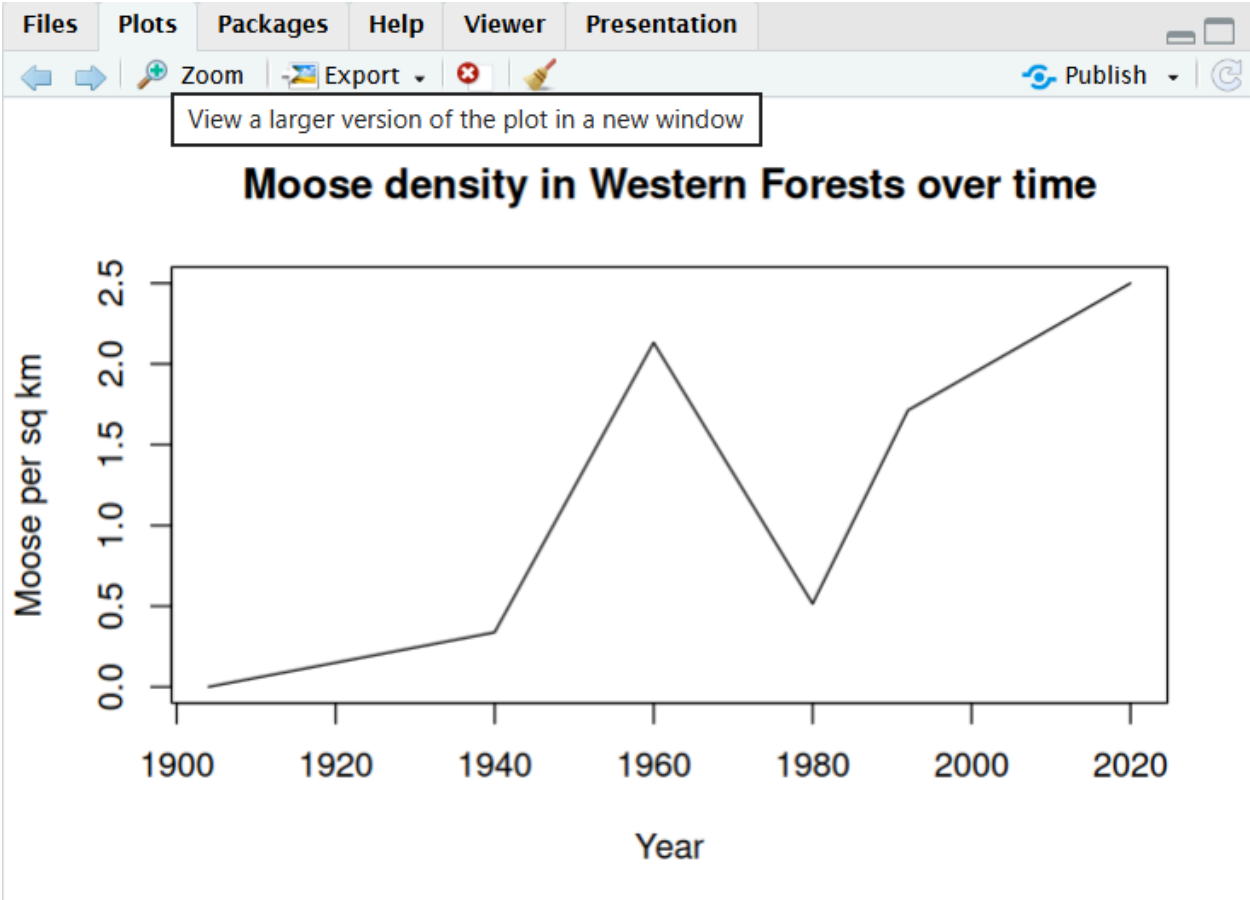
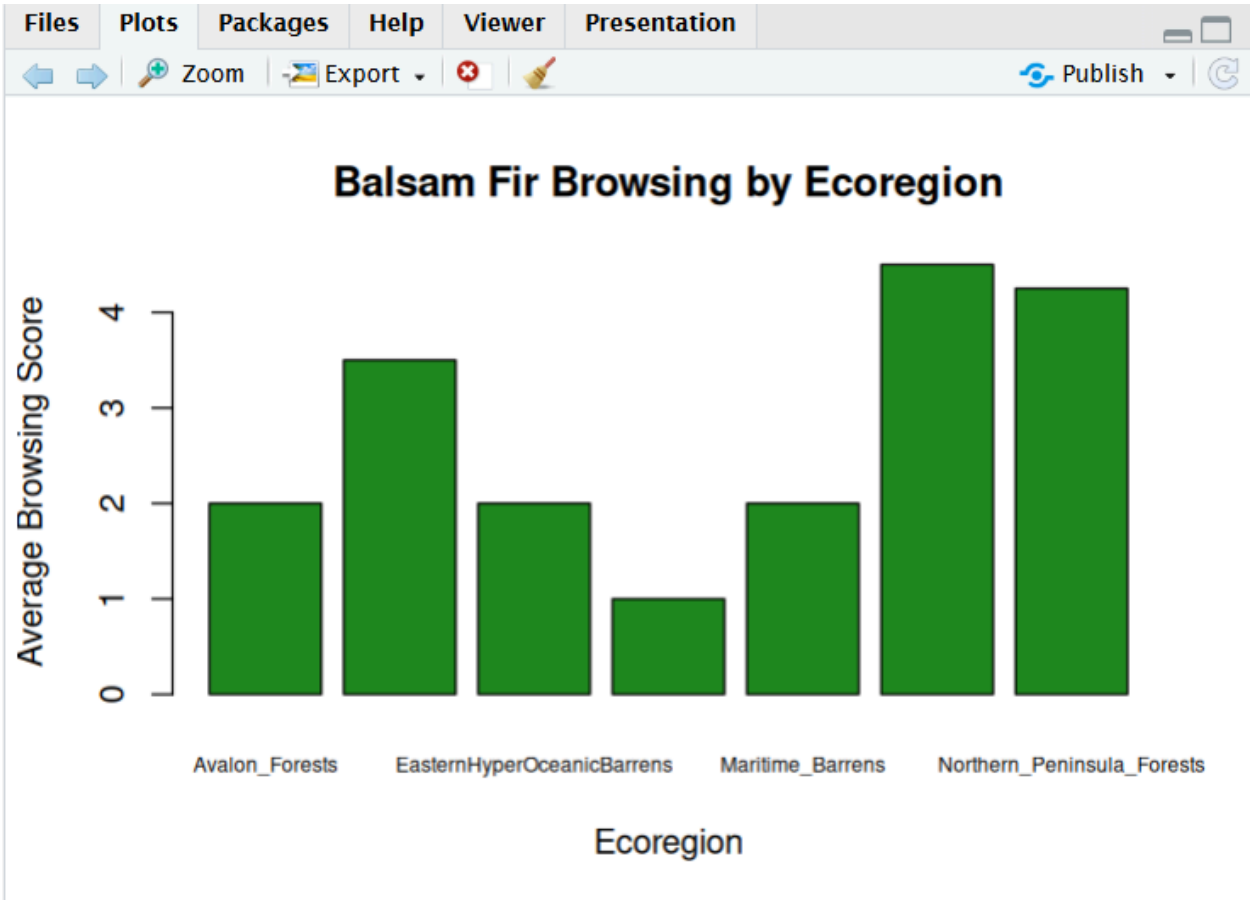

Go forward to the next source location (Ctrl+F10)

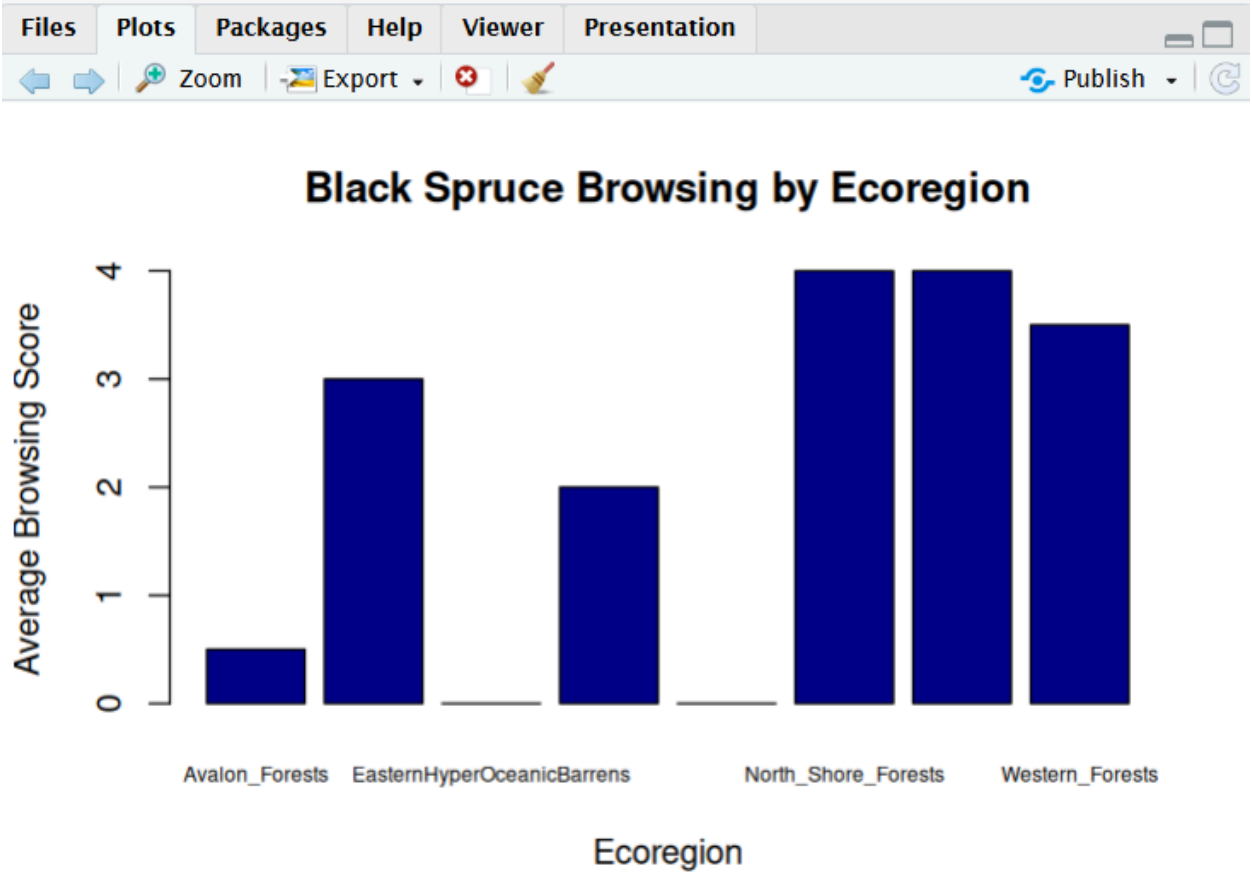
Danika McNeill
Biology 1002
11 am Slot

Biology 1002 R Assignment 2026

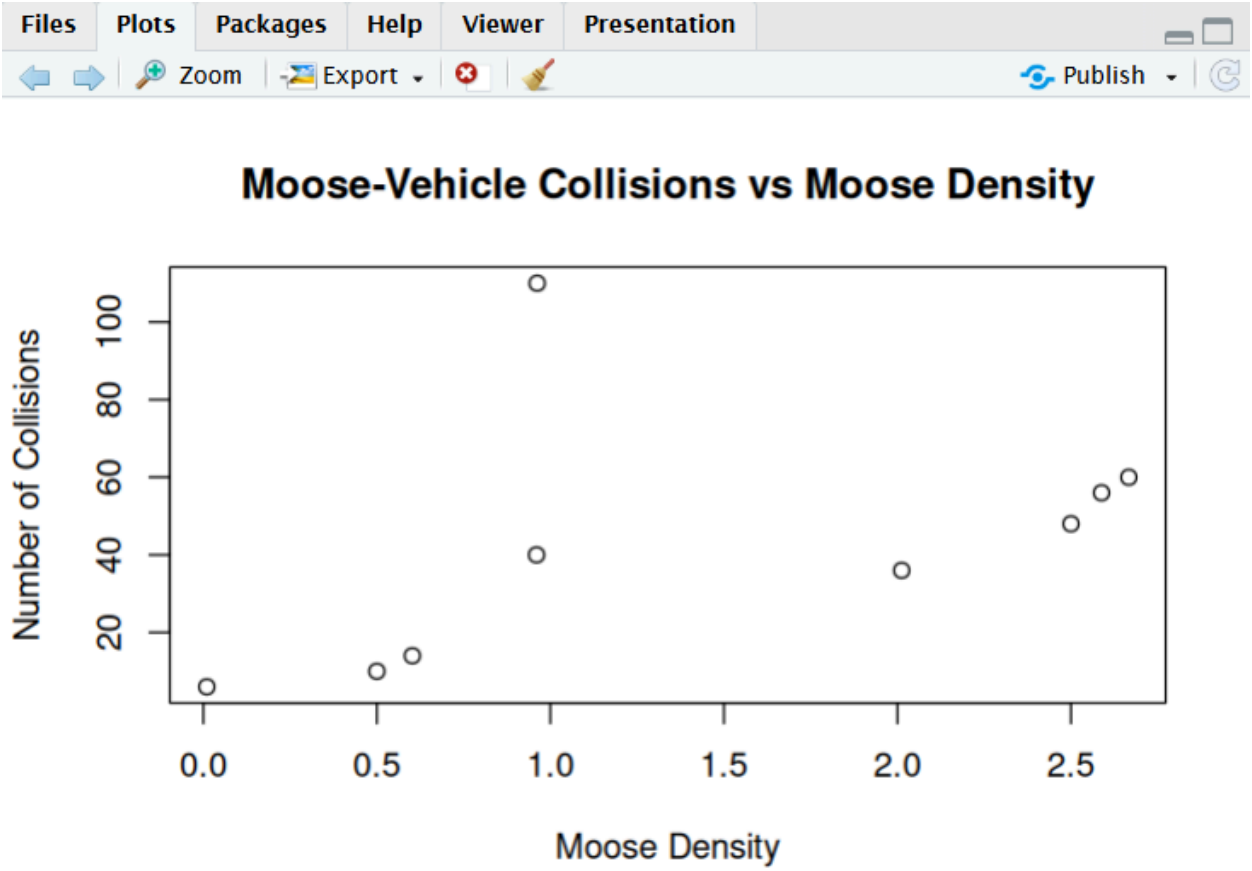


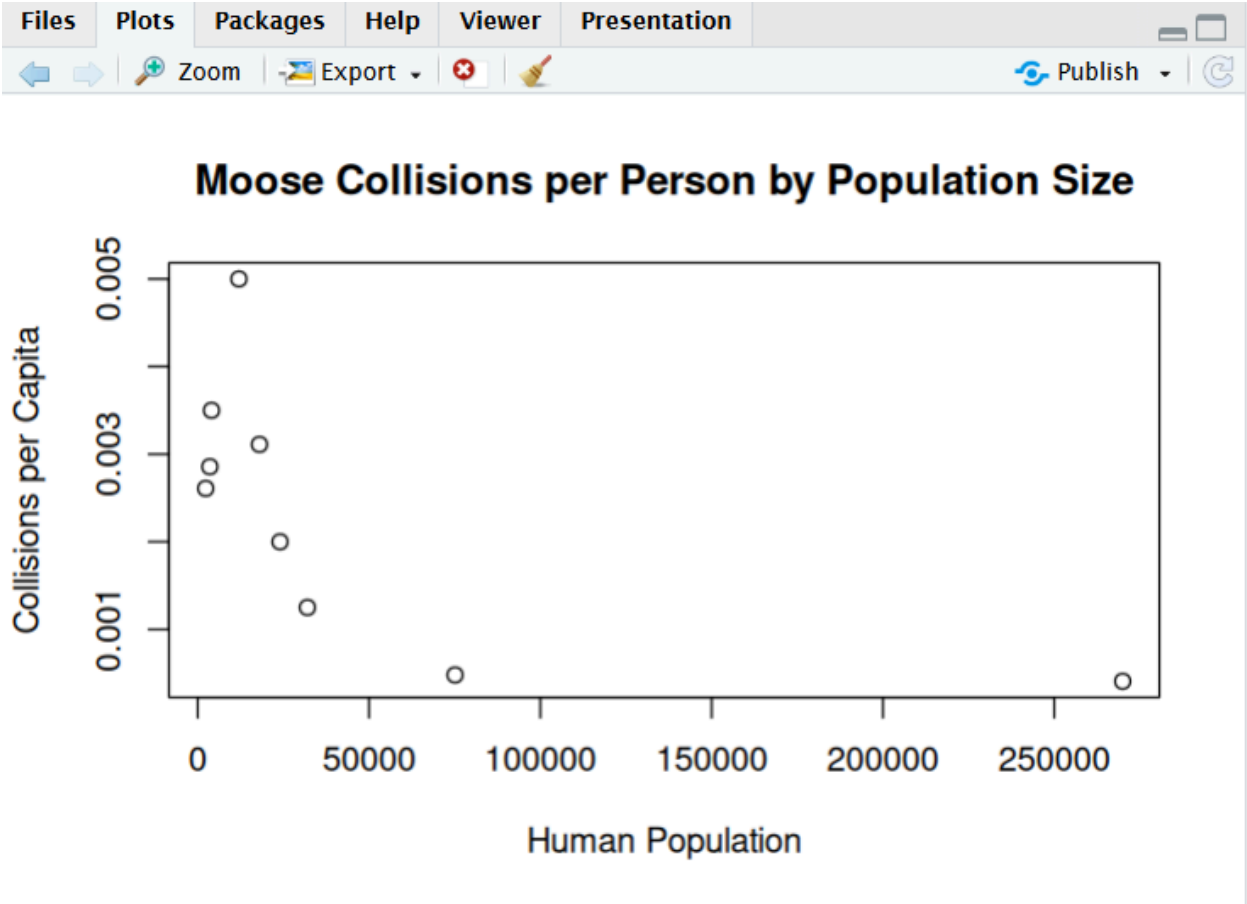

Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
Publish
View a larger version of the plot in a new window
Moose density in Western Forests over time
Moose per sq km
0.0
0.5
1.0
1.5
2.0
2.5
1900
1920
1940
1960
1980
2000
2020
Year


Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
Publish
Balsam Fir Browsing by Ecoregion
Average Browsing Score
0
1
2
3
4
Avalon_Forests
EasternHyperOceanicBarrens
Maritime_Barrens
Northern_Peninsula_Forests
Ecoregion


Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
Publish
Black Spruce Browsing by Ecoregion
Average Browsing Score
0
1
2
3
4
Avalon_Forests
EasternHyperOceanicBarrens
North_Shore_Forests
Western_Forests
Ecoregion

Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
Publish
Browsing Intensity Across Moose Density by Species
Average Browsing Score
0
1
2
3
4
5
0
1
2
Average Moose Density
Species
Alder
Balsam_Fir
Black_Ash
Black_Spruce
White_Birch
Willow


Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
Publish
Moose-Vehicle Collisions vs Moose Density
Number of Collisions
20
40
60
80
100
0.0
0.5
1.0
1.5
2.0
2.5
Moose Density




```
# # BIOL 1002 R Assignment
# Name: Danika McNeill
# Date: February 11

# FIRST: Set working directory where the CSV files are
# Example: setwd("C:/Users/Danika/Desktop")  # Windows
# Example: setwd("~/Desktop")  # Mac

# Install packages (only need to do this once)
install.packages("dplyr")
install.packages("ggplot2")

# Question 1: Load dplyr
library(dplyr)

# Question 2: Import data
```

```
moosedata <- read.csv("MoosePopulation.csv")

# Question 3: Remove missing values
moose_clean <- na.omit(moosedata)

# Question 4: Select specific columns
moose_sel <- select(moose_clean, Ecoregion, Year, Area, Estimated_Moose_Pop)

# Question 5:
# a) Oldest observation
year_min <- min(moose_sel$Year)

# b) Highest moose population
moose_max <- max(moose_sel$Estimated_Moose_Pop, na.rm = TRUE)

# Question 6: Calculate moose density
moosedata2 <- mutate(moose_sel, MooseDensity = Estimated_Moose_Pop / Area)

# Question 7: Line graph of moose density over time
plot(moosedata2$Year, moosedata2$MooseDensity,
    type = "l",
    xlab = "Year",
    ylab = "Moose per sq km",
    main = "Moose density in Newfoundland ecoregions over time")

# Question 8:
# a) Filter for Western_Forests
moose_west <- filter(moosedata2, Ecoregion == "Western_Forests")

# b) Line graph for Western Forests
plot(moose_west$Year, moose_west$MooseDensity,
    type = "l",
    xlab = "Year",
    ylab = "Moose per sq km",
    main = "Moose density in Western Forests over time")

# Question 9:
# a) Filter for year 2020
moose_2020 <- filter(moosedata2, Year == 2020)

# b) Filter for high density (> 2.0)
```

```
moose_2020_high <- filter(moose_2020, MooseDensity > 2.0)

# c) Arrange in descending order
moose_2020_high_byD <- arrange(moose_2020_high, desc(MooseDensity))

# Question 10: Use pipes for same operations
moosefinal <- moosedata2 %>%
  filter(Year == 2020) %>%
  filter(MooseDensity > 2.0) %>%
  arrange(desc(MooseDensity)) %>%
  print()

# PART II: Tree Sapling Study

# Question 11:
# a) Load sapling data
saplings <- read.csv("SaplingStudy.csv")

# b) Remove NAs
sap_clean <- na.omit(saplings)

# Question 12:
# a) Average browsing by ecoregion
sap_reg_browse <- sap_clean %>%
  group_by(Ecoregion) %>%
  summarize(AverageBrowsing = mean(BrowsingScore)) %>%
  print()

# b) Arrange and comment
avg_browse_reg <- arrange(sap_reg_browse, desc(AverageBrowsing))
# Western_Forests has highest browsing (5.0), Long_Range_Barrens has lowest (2.6)

# Question 13:
# a) Average height by ecoregion
sap_reg_height <- sap_clean %>%
  group_by(Ecoregion) %>%
  summarize(AverageHeight = mean(Height)) %>%
  print()

# b) Filter for low heights
sap_reg_height_low <- filter(sap_reg_height, AverageHeight < 20)
```

```
print(sap_reg_height_low)
# No ecoregions have average height less than 20 cm

# Question 14:
# a) Average browsing by species
sap_spe_browse <- sap_clean %>%
  group_by(Species) %>%
  summarize(AverageBrowsing = mean(BrowsingScore)) %>%
  print()

# b) Arrange and comment
avg_browse_spe <- arrange(sap_spe_browse, desc(AverageBrowsing))
# Black_Ash has highest browsing (5.0), Black_Spruce has lowest (2.4)

# Question 15: Balsam Fir browsing by ecoregion
fir_reg_browse <- sap_clean %>%
  filter(Species == "Balsam_Fir") %>%
  group_by(Ecoregion) %>%
  summarize(AverageBrowsing = mean(BrowsingScore)) %>%
  print()

# Question 16: Bar graph for Balsam Fir
barplot(fir_reg_browse$AverageBrowsing,
        names.arg = fir_reg_browse$Ecoregion,
        xlab = "Ecoregion",
        ylab = "Average Browsing Score",
        main = "Balsam Fir Browsing by Ecoregion",
        col = "forestgreen",
        cex.names = 0.6)

# Question 17:
# a) Black Spruce browsing by ecoregion
spruce_reg_browse <- sap_clean %>%
  filter(Species == "Black_Spruce") %>%
  group_by(Ecoregion) %>%
  summarize(AverageBrowsing = mean(BrowsingScore)) %>%
  print()

# b) Bar graph for Black Spruce
barplot(spruce_reg_browse$AverageBrowsing,
        names.arg = spruce_reg_browse$Ecoregion,
```

```
      xlab = "Ecoregion",
      ylab = "Average Browsing Score",
      main = "Black Spruce Browsing by Ecoregion",
      col = "darkblue",
      cex.names = 0.6)

# c) Comparison comment
# Black Spruce generally has lower browsing scores than Balsam Fir across ecoregions

# Question 18: Count trees by ecoregion
sap_reg_tally <- sap_clean %>%
  group_by(Ecoregion) %>%
  tally() %>%
  print()

# Question 19: Count trees by species
sap_spe_tally <- sap_clean %>%
  group_by(Species) %>%
  tally() %>%
  print()

# Question 20:
# a) Sampling bias comment
# The dataset is not evenly distributed. North_Shore_Forests has 8 samples while some
regions have only 5.

# b) Importance of recognizing bias
# Recognizing bias is important because uneven sampling can lead to incorrect
conclusions about species distributions and browsing patterns.

# PART III: Creating and Joining Datasets

# Question 21:
# a) Prepare moose data for 2020
moose_2020b <- moose_clean %>%
  filter(Year == 2020) %>%
  mutate(MooseDensity = Estimated_Moose_Pop / Area)

# b) Join datasets
moose_sap <- left_join(moose_2020b, sap_clean, by = 'Ecoregion')
```

```
# Question 22: Average browsing and density by species and ecoregion
sum_spe_browse <- moose_sap %>%
  group_by(Species, Ecoregion) %>%
  summarize(AvgBrowsing = mean(BrowsingScore),
            AvgDensity = mean(MooseDensity)) %>%
  print()

# Question 23:
# Load ggplot2
library(ggplot2)

# Create the graph for visualization
ggplot(sum_spe_browse, aes(x = AvgDensity, y = AvgBrowsing, color = Species)) +
  geom_point(size = 3) +
  theme_minimal() +
  labs(title = "Browsing Intensity Across Moose Density by Species",
       x = "Average Moose Density",
       y = "Average Browsing Score")

# a) Hypothesis comment
# There is limited evidence supporting the hypothesis. Browsing patterns don't clearly
change with moose density.

# b) Species preference comment
# Moose seem to favor Black_Ash and Willow most, and browse Black_Spruce the least.

# c) Missing species comment
# Black_Ash is not shown on the figure because it only appears in one ecoregion.

# Question 24: Create collisions dataset
collisions2020 <- c(56, 60, 14, 36, 48, 10, 40, 110, 6)
human_pop <- c(18000, 12000, 4000, 75100, 24000, 3500, 32000, 270000, 2300)
study_sites <- c("North_Shore_Forests","Northern_Peninsula_Forests",
"Long_Range_Barrens","Central_Forests","Western_Forests","EasternHyperOceanicB
arrens","Maritime_Barrens","Avalon_Forests","StraitOfBelleIsleBarrens")

moose_coll <- data.frame(collisions2020, human_pop, study_sites)

# Question 25:
# a) Rename column
moose_coll2 <- rename(moose_coll, Ecoregion = study_sites)
```

```
# b) Join datasets
coll_merge <- left_join(moose_2020, moose_coll2, by = "Ecoregion")

# Question 26:
# a) Scatterplot
plot(coll_merge$MooseDensity, coll_merge$collisions2020,
    xlab = "Moose Density",
    ylab = "Number of Collisions",
    main = "Moose-Vehicle Collisions vs Moose Density")

# b) Trend comment
# There appears to be a positive trend: higher moose density correlates with more
collisions. Avalon_Forests is an outlier.

# Question 27: Calculate collisions per capita
coll_merge_per_capita <- coll_merge %>%
  mutate(coll_per_capita = collisions2020 / human_pop)

# Question 28: Scatterplot of collisions per capita
plot(coll_merge_per_capita$human_pop, coll_merge_per_capita$coll_per_capita,
    xlab = "Human Population",
    ylab = "Collisions per Capita",
    main = "Moose Collisions per Person by Population Size")

# Question 29: Trend comment
# Collisions per capita decrease as human population increases, which makes sense
because larger populations dilute the risk per person.
```

R. file | R. file 2

Run | Source

```r
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# Date: February 11
# Install packages
install.packages("dplyr")
# BIOL 1002 R Assignment
# Name: Danika McNeill
# Date: February 11, 2026
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plot(moosedata2$Year, moosedata2$MooseDensity,
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ylab = "Moose per sq km",
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# a) Filter for Western_Forests
moose_west <- filter(moosedata2, Ecoregion == "Western_Forests")
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plot(moose_west$Year, moose_west$MooseDensity,
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main = "Moose density in Western Forests over time")
# Question 9:
# a) Filter for year 2020
moose_2020 <- filter(moosedata2, Year == 2020)
# b) Filter for high density (> 2.0)
moose_2020_high <- filter(moose_2020, MooseDensity > 2.0)
# c) Arrange in descending order
moose_2020_high_byD <- arrange(moose_2020_high, desc(MooseDensity))
# Question 10: Use pipes for same operations
moosefinal <- moosedata2 %>%
filter(Year == 2020) %>%
filter(MooseDensity > 2.0) %>%
arrange(desc(MooseDensity)) %>%
print()
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# Question 11:
# a) Load sapling data
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# b) Remove NAs
sap_clean <- na.omit(saplings)
# Question 12:
# a) Average browsing by ecoregion
```

```r
sap_reg_browse <- sap_clean %>%
group_by(Ecoregion) %>%
summarize(AverageBrowsing = mean(BrowsingScore)) %>%
print()
# b) Arrange and comment
avg_browse_reg <- arrange(sap_reg_browse, desc(AverageBrowsing))
# Western_Forests has highest browsing (5.0), Long_Range_Barrens has lowest (2.6)
# Question 13:
# a) Average height by ecoregion
sap_reg_height <- sap_clean %>%
group_by(Ecoregion) %>%
summarize(AverageHeight = mean(Height)) %>%
print()
# b) Filter for low heights
sap_reg_height_low <- filter(sap_reg_height, AverageHeight < 20)
print(sap_reg_height_low)
# No ecoregions have average height less than 20 cm
# Question 14:
# a) Average browsing by species
sap_spe_browse <- sap_clean %>%
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print()
# b) Arrange and comment
avg_browse_spe <- arrange(sap_spe_browse, desc(AverageBrowsing))
# Black_Ash has highest browsing (5.0), Black_Spruce has lowest (2.4)
# Question 15: Balsam Fir browsing by ecoregion
fir_reg_browse <- sap_clean %>%
filter(Species == "Balsam_Fir") %>%
group_by(Ecoregion) %>%
summarize(AverageBrowsing = mean(BrowsingScore)) %>%
```

Go forward to the next source location (Ctrl+F10)

```r
# Question 16: Bar graph for Balsam Fir
barplot(fir_reg_browse$AverageBrowsing,
names.arg = fir_reg_browse$Ecoregion,
xlab = "Ecoregion",
ylab = "Average Browsing Score",
main = "Balsam Fir Browsing by Ecoregion",
col = "forestgreen",
cex.names = 0.6)
# Question 17:
# a) Black Spruce browsing by ecoregion
spruce_reg_browse <- sap_clean %>%
filter(Species == "Black_Spruce") %>%
group_by(Ecoregion) %>%
summarize(AverageBrowsing = mean(BrowsingScore)) %>%
print()
# b) Bar graph for Black Spruce
barplot(spruce_reg_browse$AverageBrowsing,
names.arg = spruce_reg_browse$Ecoregion,
xlab = "Ecoregion",
ylab = "Average Browsing Score",
main = "Black Spruce Browsing by Ecoregion",
col = "darkblue",
cex.names = 0.6)
# c) Comparison comment
# Black Spruce generally has lower browsing scores than Balsam Fir across ecoregions
# Question 18: Count trees by ecoregion
sap_reg_tally <- sap_clean %>%
group_by(Ecoregion) %>%
tally() %>%
print()
```

```r
print()
# Question 16: Bar graph for Balsam Fir
barplot(fir_reg_browse$AverageBrowsing,
names.arg = fir_reg_browse$Ecoregion,
xlab = "Ecoregion",
ylab = "Average Browsing Score",
main = "Balsam Fir Browsing by Ecoregion",
col = "forestgreen",
cex.names = 0.6)
# Question 17:
# a) Black Spruce browsing by ecoregion
spruce_reg_browse <- sap_clean %>%
filter(Species == "Black_Spruce") %>%
group_by(Ecoregion) %>%
summarize(AverageBrowsing = mean(BrowsingScore)) %>%
print()
# b) Bar graph for Black Spruce
barplot(spruce_reg_browse$AverageBrowsing,
names.arg = spruce_reg_browse$Ecoregion,
xlab = "Ecoregion",
ylab = "Average Browsing Score",
main = "Black Spruce Browsing by Ecoregion",
col = "darkblue",
cex.names = 0.6)
# c) Comparison comment
# Black Spruce generally has lower browsing scores than Balsam Fir across ecoregions
# Question 18: Count trees by ecoregion
sap_reg_tally <- sap_clean %>%
group_by(Ecoregion) %>%
tally() %>%
print()
```

```r
# Question 19: Count trees by species
sap_spe_tally <- sap_clean %>%
group_by(Species) %>%
tally() %>%
print()
# Question 20:
# a) Sampling bias comment
# The dataset is not evenly distributed. North_Shore_Forests has 8 samples while some regions have on
# b) Importance of recognizing bias
# Recognizing bias is important because uneven sampling can lead to incorrect conclusions about speci
# PART III: Creating and Joining Datasets
# Question 21:
# a) Prepare moose data for 2020
moose_2020b <- moose_clean %>%
filter(Year == 2020) %>%
mutate(MooseDensity = Estimated_Moose_Pop / Area)
# b) Join datasets
moose_sap <- left_join(moose_2020b, sap_clean, by = 'Ecoregion')
# Question 22: Average browsing and density by species and ecoregion
sum_spe_browse <- moose_sap %>%
group_by(Species, Ecoregion) %>%
summarize(AvgBrowsing = mean(BrowsingScore),
AvgDensity = mean(MooseDensity)) %>%
print()
# Question 23:
# a) Hypothesis comment
# The hypothesis is not strongly supported - moose show consistent preferences across different densi
# b) Species preference comment
# Moose seem to favor Black_Ash and Willow most, and browse Black_Spruce the least.
# c) Missing species comment
# Black_Ash is not shown on the figure because it only appears in one ecoregion.
```

```
# Question 24: Create collisions dataset
collisions2020 <- c(56, 60, 14, 36, 48, 10, 40, 110, 6)
human_pop <- c(18000, 12000, 4000, 75100, 24000, 3500, 32000, 270000, 2300)
study_sites <- c("North_Shore_Forests","Northern_Peninsula_Forests", "Long_Range_Barrens","Central_Fo
moose_coll <- data.frame(collisions2020, human_pop, study_sites)
# Question 25:
# a) Rename column
moose_coll2 <- rename(moose_coll, Ecoregion = study_sites)
# b) Join datasets
coll_merge <- left_join(moose_2020, moose_coll2, by = "Ecoregion")
# Question 26:
# a) Scatterplot
plot(coll_merge$MooseDensity, coll_merge$collisions2020,
xlab = "Moose Density",
ylab = "Number of Collisions",
main = "Moose-Vehicle Collisions vs Moose Density")
# b) Trend comment
# There appears to be a positive trend: higher moose density correlates with more collisions. Avalon_
# Question 27: Calculate collisions per capita
coll_merge_per_capita <- coll_merge %>%
mutate(coll_per_capita = collisions2020 / human_pop)
# Question 28: Scatterplot of collisions per capita
plot(coll_merge_per_capita$human_pop, coll_merge_per_capita$coll_per_capita,
xlab = "Human Population",
ylab = "Collisions per Capita",
main = "Moose Collisions per Person by Population Size")
# Question 29: Trend comment
# Collisions per capita decrease as human population increases, which makes sense because larger popu
savehistory("/cloud/project/R. Code")
```